



Indigenous
Data Network

Indigenous Data Governance as a Foundation of Global Data Governance

*Indigenous Data Network Submission to the UN CSTD
Working Group on Data Governance*

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Executive Summary

This submission is made by the Indigenous Data Network (IDN), a national Indigenous-led initiative based at the University of Melbourne that works with Aboriginal and Torres Strait Islander peoples, Indigenous communities of origin, and Indigenous knowledge holders. Established in 2018, the IDN advances Indigenous leadership, authority, and governance over data relating to peoples, lands, cultures, and knowledge systems, and collaborates with governments and research infrastructures to embed governance into operational data systems.

This submission is intended to be read alongside parallel Indigenous contributions, including the submission from Sarawak, Malaysia. Together, these submissions foreground Indigenous Data Governance as a distinct and necessary component of global data governance. While the Sarawak submission demonstrates how Indigenous governance is grounded in living customary law and sub-national institutions, this submission focuses on how governance can be implemented technically and institutionally within national and international data infrastructures. Taken together, they show that Indigenous Data Governance is both normatively grounded and operationally feasible across diverse jurisdictions.

We advance three arguments for the UN CSTD Working Group on Data Governance:

1. Indigenous Data Governance addresses a structural gap in global frameworks that assume data subjects are either individuals or states. These assumptions fail where data is collective, intergenerational, place-based, and governed through Indigenous authority.
2. Indigenous Data Governance should be treated as a cross-cutting governance layer, comparable to privacy, security, and data integrity, rather than a sectoral or ethical add-on.
3. Governance can travel with data without undermining interoperability or innovation, through machine-readable metadata, catalogue profiles, consent signalling, and auditable permissions aligned with CARE alongside FAIR.

The submission responds to the survey's four tracks, fundamental principles, interoperability, sharing benefits, and data flows, positioning Indigenous Peoples as governing authorities in data systems. Aligning with the *United Nations Declaration on the Rights of Indigenous Peoples*¹ and recent work of the *Expert Mechanism on the Rights of Indigenous Peoples on the right to data*², we call for Indigenous Data Governance to be recognised as an essential component of inclusive and trustworthy global data governance.

¹ https://www.un.org/development/desa/Indigenouspeoples/wp-content/uploads/sites/19/2018/11/UNDRIP_E_web.pdf

² <https://www.ohchr.org/en/documents/thematic-reports/ahrc6066-right-indigenous-peoples-data-including-regard-data-collection>



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PART I: Indigenous Data Governance as a Foundation of Global Data Governance

Core problem

Global data governance largely presumes that authority over data can be settled by individual consent, institutional ownership, or national jurisdiction. These presumptions fail when data is collective, intergenerational, place-based, and governed through Indigenous law and cultural authority. The outcome is the routine exclusion of Indigenous Peoples from decisions about data that materially affect their lives, lands, and futures, and a system-level gap in international data governance. This exclusion undermines self-determination, perpetuates deficit-based narratives, and limits communities' ability to use data to support their own governance, development, and priorities.

Indigenous Data Governance as a structural gap in global data governance

Contemporary global data governance frameworks are built on underlying assumptions that data subjects are either individuals or states, and that authority over data can be resolved through privacy law, contractual arrangements, or national jurisdiction. These assumptions do not hold in contexts where data is collectively held, intergenerational, place-based, and governed by Indigenous legal and cultural authority.

Indigenous Data Governance exposes a structural gap in global data governance: the absence of mechanisms to recognise and operationalise collective authority that does not align neatly with state boundaries or individual consent models. This gap is not confined to Indigenous contexts. It is increasingly visible in domains such as environmental data, cultural heritage, genetic resources, and community-based knowledge systems, where governance cannot be reduced to individual rights or institutional ownership.

Addressing Indigenous Data Governance is therefore not a marginal or sector-specific concern. It is a test case for whether global data governance frameworks can accommodate plural forms of authority, governance, and responsibility. Failure to do so risks entrenching extractive data practices, undermining trust in data systems, and weakening the legitimacy of international data governance arrangements.

Key principles



This submission is grounded in the following principles:

Recognising Indigenous Data Assets: Institutions must identify and acknowledge Indigenous data assets and Indigenous-held data, respect Indigenous custodianship, and ensure that data is findable and accessible to Indigenous communities.

Partnering with Indigenous People and Entities: Ethical governance requires genuine partnerships that centre Indigenous leadership, priorities, and decision-making. Co-design, shared governance, and long-term relationships are essential.

Building and Enabling Data Capabilities: Indigenous communities must be supported to strengthen their capacity to manage, interpret, and govern data. This includes training, infrastructure, and institutional reform.

Supporting Indigenous Self-Determination in Data Access and Use: Indigenous communities must have the ability to access and utilise data in a manner that aligns with their priorities and governance protocols.

Building an Inclusive Data Ecosystem: Institutions must foster cultural safety, transparency, and accountability across all levels of data practice, embedding Indigenous knowledge systems, where appropriate, and governance protocols.

Accordingly, this submission invites the Working Group to recognise Indigenous Data Governance as a foundational component of global data governance; to position Indigenous Peoples as governing authorities within data systems; and to advance governance-first approaches to interoperability, benefit-sharing, and data flows that allow authority, consent, and accountability to travel with data across institutional and national boundaries.

Context

Indigenous data governance is situated within United Nations mandates on Indigenous rights, self-determination, and ethical data use. This alignment reflects a conscious and sustained engagement with UN norms, which have played a formative role in shaping Indigenous data governance approaches in the Australian context. The approach outlined in the Australian Government's *Framework for the governance of Indigenous data* (NIAA, 2023) aligns directly with the United Nations mandate on Indigenous Peoples' rights to data, as articulated through the UN Declaration on the Rights of Indigenous Peoples (UNDRIP).

UNDRIP affirms that Indigenous peoples have the right to self-determination (Article 3) and the right to maintain, control, protect, and develop their cultural heritage, traditional knowledge, and cultural expressions, including data and digital knowledge systems (Article 31). This mandate is reinforced



by the United Nations Expert Mechanism on the Rights of Indigenous Peoples (EMRIP)³, which recognises data as a fundamental human right and emphasises Indigenous Peoples' authority over the collection, ownership, and use of data that concerns them. The United Nations Permanent Forum on Indigenous Issues (UNPFII)⁴ further highlights the central role of Indigenous-led data in visibility, policy development, and the exercise of rights across health, culture, environment, and governance.

Consistent with these mandates, this Framework's emphasis on Indigenous data governance, culturally grounded authority, and community decision-making reflects international standards such as the CARE Principles for Indigenous Data Governance⁵, which draw explicitly on UNDRIP and affirm collective benefit, authority, responsibility, and ethics in all data practices. In the Australian context, Indigenous data governance gives practical effect to UNDRIP principles of self-determination, cultural protection, and free, prior and informed consent (FPIC), contributes to the advancement of the Sustainable Development Goals through improved ethical data practices, aligns with UN mandates on research and cultural rights, and foregrounds the limitations of state-centred data systems in meeting international human rights obligations.



Figure 1. Relationship between the FAIR and CARE Principles for Data Governance

Source: Global Indigenous Data Alliance (GIDA); see also Carroll et al., 2020

The CARE Principles add a governance and rights-based layer to the FAIR Principles, addressing collective benefit, authority, responsibility, and ethics in the reuse of data relating to Indigenous peoples. As illustrated in Figure 1, Indigenous Data Governance is not addressed by technical data

³Study: <https://www.ohchr.org/en/documents/thematic-reports/ahrc6066-right-indigenous-peoples-data-including-regard-data-collection>

⁴ Overview: <https://social.desa.un.org/issues/indigenous-peoples/unpfii>

⁵ See: <https://doi.org/10.5334/dsj-2020-043>



governance principles alone, but requires explicit attention to authority, consent, responsibility, and collective benefit throughout the data lifecycle.

Definitions and Conceptual Framing

The following section defines key terms and conceptual propositions that inform this submission and its approach to Indigenous Data Governance for this submission.

Indigenous Data

Indigenous data encompasses a broad and evolving category of information that reflects the lived experiences, knowledge systems, and governance priorities of Aboriginal and Torres Strait Islander peoples. In contrast to narrow statistical or administrative conceptions of data, Indigenous data is understood as relational, contextual, and governed according to Indigenous laws, values, and responsibilities. The Indigenous Data Network (IDN) defines Indigenous data as:

Data generated, intentionally or not, by, about, or for Aboriginal and Torres Strait Islander people. Indigenous data refers to information, in any format or medium, collected, analysed, stored, and interpreted within the context of Indigenous individuals, collectives, populations, entities, lifeways, cultures, knowledge systems, lands, biodiversity, water and other resources. It includes data collected, used, or stored by any agency, department, laboratory, organisation, corporation, statutory body, university or research institute, conducted by, with, and about Indigenous people or people, and data that Indigenous communities have generated and maintained themselves. (Indigenous Data Network, 2024)

This definition highlights several interrelated dimensions that are central to Indigenous data governance:

Scope: Indigenous data is not limited to demographic or statistical information. It includes cultural, ecological, linguistic, genealogical, and community-generated knowledge, whether digitised or held in oral, visual, or material forms.

Contextuality: The meaning and significance of Indigenous data are shaped by Indigenous worldviews, relationships, and responsibilities. Data must be understood within the cultural, historical, and political contexts in which it is produced and used.

Multiplicity of sources: Indigenous data may be held by governments, research institutions, corporations, or Indigenous organisations. It includes both externally generated data and data created and maintained by Indigenous communities themselves.

Relational Ownership and Governance: Indigenous data is not simply about representation, but about authority, access, accountability, and decision-making. Indigenous communities have the right



to determine how data relating to them is collected, interpreted, accessed, linked, and used, in accordance with their own governance structures.

This definition is both expansive and relational. It underpins ethical and culturally appropriate data practices and supports Indigenous data governance by restoring authority and agency to Indigenous people over the data that affects their lives and futures.

In the Australian context, it is also important to recognise that Indigenous peoples are implicated in, and affected by, all national data systems, despite forming a minority within the general population. This underscores the limitations of population-based data models and the need for governance approaches that address Indigenous interests across the data ecosystem.

Indigenous Data Sovereignty

The concept of Indigenous data sovereignty has emerged as a powerful articulation of Indigenous peoples' rights to control data relating to their communities, cultures, lands, and knowledges. It is grounded in broader political and cultural struggles for self-determination and the reassertion of authority over Indigenous knowledge systems (Kukutai & Taylor, 2016). In Australia, the concept has gained traction in response to the historical and ongoing extraction, misrepresentation, and mismanagement of Indigenous data by state and institutional actors (Rose, Langton, Smith, & Clinch, 2023).

The language of sovereignty, while politically resonant, remains conceptually and legally complex. In contexts such as Australia, Indigenous sovereignty is not formally recognised by the state and therefore lacks enforceable legal standing. This creates a persistent tension: while Indigenous communities may assert sovereignty over data as a matter of rights and self-determination, state and institutional systems often lack mechanisms capable of recognising or upholding those claims in practice (Carroll et al., 2019).

These tensions are intensified when sovereignty is applied to data governance. Sovereignty, as developed within Western political and legal traditions, is closely associated with territorial authority, jurisdictional exclusivity, and bounded state power. Data, by contrast, is inherently non-territorial. It moves across institutional and national boundaries, is frequently replicated and linked, and circulates within complex digital infrastructures. For Indigenous peoples situated within nation-states asserting overarching sovereignty, these dynamics complicate claims to exclusive authority over data and challenge conventional notions of possession and control central to sovereignty discourse.

The concept of sovereignty also carries a contested historical legacy in Indigenous contexts. It was mobilised to legitimise colonial expansion and the dispossession of Indigenous lands and governance systems, often negating pre-existing Indigenous authority. Contemporary Indigenous engagements with sovereignty discourse may therefore be understood as strategic and pragmatic attempts to translate Indigenous rights claims into terms legible within dominant legal frameworks. At the same time, sovereignty framings can sit uneasily with Indigenous worldviews that prioritise relationality, stewardship, and collective responsibility over absolute authority or territorial control.

In practice, Indigenous data sovereignty **is** best understood as a form of qualified or relational authority: a negotiated expression of Indigenous rights and governance that operates within, across,



and sometimes in tension with state-centred legal and institutional systems. While sovereignty remains a vital political and ethical claim, its limitations in operational and institutional settings underscore the need for complementary frameworks capable of translating Indigenous authority into practicable governance arrangements.

Indigenous Data Governance

Indigenous data governance provides a pragmatic and culturally grounded framework for addressing the complex challenges of data management in Indigenous contexts. Rather than focusing on abstract claims of authority alone, data governance centres on the practical work of decision-making, control, access, use, and accountability across the data lifecycle. The concept of governance offers the flexibility required to encompass diverse Indigenous laws, values, and practices, while enabling engagement with existing legal, policy, and institutional systems.

As a framework, governance resonates with Indigenous concepts of stewardship, responsibility, and collective decision-making. As a relational and practice-oriented framework, it supports collaborative approaches without relinquishing Indigenous authority or cultural integrity. Framing Indigenous data rights through governance fosters more productive engagement with governments, institutions, and other actors by emphasising implementation, accountability, and shared responsibility rather than abstract or binary legal recognition. This enables Indigenous communities to address concrete challenges, including access to stranded data held by government agencies, navigation of jurisdictional ambiguity within non-Indigenous institutions, and the embedding of Indigenous epistemologies within metadata, classification, and archival systems (Kukutai & Taylor, 2016; Carroll et al., 2019; Rose et al., 2023).

Indigenous data governance does not replace sovereignty: it operationalises it. Sovereignty, as a concept, is frequently mediated through state recognition and presupposes a binary relationship of recognition or denial. Governance, by contrast, is relational, iterative, and adaptive. It provides the mechanisms through which Indigenous communities can enact authority over data in practice, even in the absence of formal legal recognition. In this way, governance enables context-specific negotiations of authority grounded in Indigenous law, values, and relational worldviews, while facilitating collaboration with researchers, institutions, and governments on Indigenous-defined terms.

For international data governance processes, including those led by the United Nations, Indigenous data governance offers a more institutionally actionable framework than sovereignty alone. Governance translates rights into decision-making roles, operational rules, and technical controls that can be embedded within data infrastructures, standards, and cross-border arrangements. Indigenous data governance is therefore not external to global data governance agendas; it is a necessary extension of them.

Indigenous data governance is not monolithic. It takes diverse legal, cultural, and institutional forms across regions, including settings where Indigenous customary law holds formal legal authority, and others where governance must be negotiated within state-centric systems. Global data governance frameworks must be capable of recognising and accommodating this plurality rather than imposing



a singular model. This diversity strengthens, rather than weakens, the case for Indigenous data governance as a foundational component of international data governance.

Key Challenges

Indigenous peoples face persistent challenges across local, regional, and global data ecosystems in which information about their lives, lands, and cultures is predominantly generated, held, and controlled by governments and institutions rather than by Indigenous communities. As a result, Indigenous peoples are frequently rendered invisible or misrepresented in national data systems and are excluded from decision-making about how data is collected, classified, analysed, and used. Indigenous control over data is therefore a fundamental aspect of self-determination and a necessary response to long-standing patterns of extraction, misrepresentation, and institutional imbalance. These challenges underscore the importance of distinguishing between different data regimes and the governance assumptions that underpin them.

A central challenge across all data regimes is that Indigenous governance does not travel with data. As data moves across systems, sectors, and jurisdictions, authority, consent conditions, and cultural protocols are routinely stripped from records, treated as contextual rather than infrastructural elements of data systems. This leads to downstream use and reuse that is disconnected from Indigenous governance, even where data was initially collected under Indigenous-specific agreements or ethical approvals.

Further, prevailing reliance on individual consent models remains poorly suited to Indigenous contexts. Individual, transactional consent mechanisms fail to account for collective, intergenerational, and place-based Indigenous authority, creating governance gaps even where consent procedures formally exist. These limitations are particularly acute in linked, administrative, and large-scale datasets, where decisions about reuse and secondary analysis are detached from the circumstances of original consent.

These challenges are further compounded by jurisdictional fragmentation. Indigenous data routinely traverses institutional, sectoral, and national boundaries, often through linkage, interoperability, and cross-border data flows. In the absence of mechanisms capable of recognising Indigenous authority across jurisdictions, data movement can further erode governance, accountability, and community oversight.

Within this broader context, different data regimes present distinct risks and constraints:

Open Data: Open data models promote transparency and public access by making data freely available for use and reuse. While such models support accountability and innovation, they can amplify risks for Indigenous peoples when governance conditions are absent. In particular, open data can facilitate extraction, misinterpretation, and misuse of Indigenous data, especially where cultural context, collective authority, and consent are not recognised.



State-Owned Data: Data collected and held by government agencies (e.g. health, education, census, and welfare systems) rarely recognises Indigenous authority or community-defined protocols. These data are typically treated as state assets and governed through national regulatory frameworks and institutional policies, with limited mechanisms for Indigenous participation in governance, access decisions, or downstream use.

Corporate Data: Corporate data practices largely prioritise commercial and proprietary interests. They generally operate without obligations to recognise Indigenous collective rights, cultural responsibilities, or community-defined conditions for data use, creating heightened risks of unaccountable extraction and inequitable benefit-sharing.

Indigenous-Controlled Data: While Indigenous-controlled data offers the greatest alignment with Indigenous governance principles, it is frequently constrained by limited and uneven investment in infrastructure, capability, and long-term resourcing. Indigenous content is also commonly unrecognised or inconsistently classified across institutional systems, limiting interoperability and visibility within broader data ecosystems.

These challenges highlight the need for data governance approaches that recognise Indigenous authority across all data regimes, rather than treating Indigenous data as a marginal subset within state, corporate, or open data systems. These distinctions have direct implications for global data governance, particularly in relation to interoperability, cross-border data flows, and the design of data infrastructures that claim to be inclusive or rights-based.

Together, these challenges point to the need for governance-first architectures in which authority, consent, and cultural protocols are expressed as machine-readable metadata and portable rules, so governance travels with data across systems and borders.



PART II: Operationalising Indigenous Data Governance — Integrated Responses to the UN CSTD Survey (Tracks 1–4)

This section provides integrated responses to the Working Group's survey across Tracks 1–4; it should be read in conjunction with Part I, which sets out the governance rationale and definitions that underpin the recommendations.

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Track 1. Fundamental principles of data governance

Q6. What additional data governance frameworks should be included in the Secretariat's compilation?

Indigenous data governance should be recognised as a distinct domain of data governance, rather than subsumed within research ethics frameworks. While research ethics address project-level conduct, they are insufficient for governing administrative, statistical, and linked data systems that operate at institutional and national scales.

Operationally, Indigenous governance must be able to travel with data. This requires governance to be embedded within data systems through catalogue standards and machine-readable permissions, so that authority, consent status, sensitivity, and permitted or prohibited uses remain visible and auditable as data moves across systems and jurisdictions (United Nations General Assembly, 2007; EMRIP, 2025; W3C, 2024; Iannella et al., 2018).

Clear articulation of the distinct roles of ownership, custodianship, and stewardship further clarifies who holds authority to decide, who stores and provides access, and who may process, link, or derive value from data, and under what conditions (Rose et al., 2023).

Q7. Which principles are particularly important?

Fundamental principles include recognition of Indigenous Peoples as governing authorities over data relating to them; collective, ongoing free, prior and informed consent (FPIC); governance before access; limits to openness, including refusal rights; and auditable accountability across the data lifecycle. These principles align with United Nations human rights instruments and are complementary to both the FAIR and CARE Principles (United Nations General Assembly, 2007; Carroll et al., 2020; Wilkinson et al., 2016). Governance should also address collective and cultural harms that may arise even where individual privacy protections are met, recognising that harms can be community-wide, intergenerational, and context-dependent.

Recognition of Indigenous authority, collective and ongoing FPIC, refusal rights, and auditable accountability should be treated as minimum requirements for rights-based data governance rather than optional good practice.

Q8. Are there sector-specific nuances the Track ought to highlight?

Data is not neutral. For Indigenous Peoples, data is embedded in lifeways, languages, and responsibilities. Governance therefore needs to be relational, linking Indigenous communities of



origin and Indigenous knowledge holders with institutions and downstream users, so that meaning, context, and obligations travel with the record rather than being detached at the point of reuse (Smith & Clarke, 2025).

Accordingly, data governance must be portable and machine-readable. Governance should not be confined to static agreements or policy documents but expressed within catalogue metadata so that authority, consent status, sensitivity, and community-defined protocols remain visible wherever records circulate. In this respect, the CARE Principles complement FAIR by making people, purpose, and power legible within technical systems (Smith & Clarke, 2025).

These positions align with EMRIP's analysis of the right to data and the need to centre Indigenous worldviews within data system design (EMRIP, 2025).

From a systems perspective, governance visibility should be treated as a design requirement for interoperable data infrastructures, not as external documentation separate from technical workflows.

Q9. Which tensions or trade-offs among the proposed data governance principles do you consider most pressing in your context, and what mechanisms would you recommend to resolve or balance these tensions?

Key tensions include openness versus collective rights, efficiency versus consent, and individual privacy versus collective harms. These tensions can be addressed through governance-first architectures, trusted or federated environments that bring computation to data, and machine-readable permissions that regulate downstream use and reuse (Lin et al., 2020; W3C, 2024; Iannella et al., 2018).

Q10. Are there any other points, ideas, or issues you would like to raise that were not addressed in this survey?

Indigenous Peoples should be positioned as continuing data governors within national statistical systems and multilateral data infrastructures, with decision-making roles rather than stakeholder or advisory status, consistent with United Nations human rights obligations and EMRIP guidance (United Nations General Assembly, 2007; EMRIP, 2025).

Track 2. Interoperability

Q7. What are the key challenges and risks to achieving interoperability at the national, regional, and international levels?

Key challenges include layered authority across Indigenous, sub-national, national, and international contexts; variation in governance arrangements at regional and community levels; capital-city bias



in system design; and heightened risks of extraction when technical interoperability advances faster than governance frameworks. Where alignment focuses on data standards alone, without corresponding governance mechanisms, interoperability can amplify rather than mitigate harm (EMRIP, 2025; Rose et al., 2023).

Q8. In which areas of development could interoperability have the most significant positive impact for your area of work, country or region in the next decade?

Interoperability has significant potential where data linkage is necessary for policy and service improvement, including administrative and health data; justice, child and family, housing, and education datasets; and environmental and cultural heritage data. In these domains, interoperability must carry governance and language metadata so that authority, context, and Indigenous governance requirements remain intact across systems (Wilkinson et al., 2016; Lin et al., 2020).

A persistent barrier to effective interoperability is the inconsistent recognition of Indigenous content within organisations. Data relating to Indigenous peoples is often unidentified, classified according to differing criteria, or recorded inconsistently across systems. A standardised, transparent approach to identifying and recording Indigenous data would materially improve interoperability, governance visibility, and accountability.

Q9. How can interoperability be advanced in ways that support collaboration while protecting each country's control over its own data, systems, and development priorities?

Interoperability can be advanced through governance-first agreements, trusted or federated environments, and catalogue records that carry provenance, authority, consent status, sensitivity, and permitted or prohibited uses. These approaches enable collaboration while allowing countries and communities to retain control over their data and governance settings (W3C, 2024; Iannella et al., 2018).

Adopting shared reference data and standardised assessment and classification processes further supports interoperability. Reference data platforms such as the *IDN Reference Data Catalogue*⁶ provide common points of alignment that can be reused and extended across institutions, promoting shared understanding of Indigenous data-related concepts without centralising control.

Q10. How can technical and semantic standards be developed or adapted so they work across different cultural, linguistic, and institutional contexts?

⁶ <https://data.idnau.org/catalogs>



Within the *Improving Indigenous Research Capabilities*⁷ initiative, governance-first interoperability is implemented through infrastructure. An IDN Catalogue Profile, developed as an extension of DCAT, introduces Indigenous-specific governance fields and mappings; the Metadata Entry Tool prompts FAIR- and CARE-aligned description at the point of cataloguing; and a knowledge graph records relationships, so decision rights remain traceable across catalogues and time (Smith & Clarke, 2025).

Many standards already support multiple languages and contexts. Glossaries of Indigenous data governance terms can be presented in different languages and structured for different user groups while retaining shared reference definitions. Supporting materials can similarly be tailored for technical and non-technical audiences without altering underlying models. Profiling mechanisms (e.g. formalised specialisations of existing standards) allow Indigenous-specific requirements to be met while maintaining conformance with general-purpose standards such as DCAT.

Q11. How can governance and legal frameworks be designed to enable cooperation on interoperability without creating reliance on external actors or systems and remain aligned with public interest?

Governance frameworks should be transparent about the definitions, methods, and reference resources they use. Processes for identifying and classifying data as Indigenous should be publicly documented, based on clear criteria and shared reference definitions, enabling consistent application across institutions and jurisdictions.

Q12. What approaches could make interoperability efforts financially and institutionally sustainable over time?

Sustainability can be supported through cost-sharing models that prioritise reuse and extension of existing reference resources rather than duplication. Interoperability efforts should also leverage established, non-Indigenous data tooling and practices, such as profiling and modular standards, adapted to Indigenous governance requirements, reducing long-term maintenance costs and institutional burden.

Q13. What capacities, skills, or partnerships are most important to strengthen the ability to manage and govern interoperable systems?

Effective interoperability requires skills in data curation akin to collections management and librarianship, as well as foundational data modelling and classification expertise. Understanding how vocabularies and taxonomies function, including hierarchical and poly-hierarchical structures, is

⁷ <https://ardc.edu.au/project/improving-indigenous-research-capabilities/>



critical for accurate metadata creation and governance signalling, particularly for Indigenous reference resources such as the Indigenous Themes vocabulary.⁸

Q14. Which interoperability framework at the national, regional, or international level might be considered a good practice?

Good practice combines governance and infrastructure, including contractual clauses embedding collective consent and remedies; Indigenous intermediaries; recurrent investment; shared registries; and partnerships that support federated models of access. Technical frameworks such as DCAT⁹ extended with CARE and FAIR principles, alongside Five Safes, ALCOA+, and trusted environments, provide practical foundations for governed interoperability (EMRIP, 2025; Lin et al., 2020; W3C, 2024; Desai et al., 2016; World Health Organization, 2021).

While Australia has not yet formally designated Indigenous-specific interoperability frameworks, candidates are emerging. The *IDN Catalogue Profile*¹⁰ builds on established best-practice cataloguing standards and demonstrates how Indigenous governance requirements can be embedded within general-purpose systems rather than treated as exceptions.

Treating governance visibility as a design requirement for interoperable systems is necessary to prevent collective and cultural harms, aligning technical data exchange with United Nations human-rights obligations.

Track 3. Sharing the Benefits of Data

Q8. What dimensions of “benefits” (economic, social and other indirect aspects of it) should the track consider and why?

Benefits associated with Indigenous data must be understood as multi-dimensional and community-defined, extending beyond economic value to encompass cultural, social, environmental, and governance outcomes. Critically, benefits should not be assumed or fixed in advance but defined by Indigenous communities and revisited across the data lifecycle as contexts, risks, and priorities change (Carroll et al., 2020; Rose et al., 2023). This approach recognises that value emerges through governed use, rather than as an inherent property of data.

⁸ IDN Themes vocabulary: <https://data.idnau.org/pid/vocab/idn-th>

⁹ World Wide Web Consortium (2023). Data Catalog Vocabulary (DCAT). International Standard. <https://www.w3.org/TR/vocab-dcat-3/>

¹⁰ <https://idnau.org/resources/catalogprofile>



Q9. How do stakeholders involved throughout the data lifecycle (e.g., collection, curation, analysis, and reuse) impact the benefits of data?

Decisions made at each stage of the data lifecycle, including collection, description, access, analysis, sharing, and reuse, shape whether data delivers benefit or harm. Governance visibility is therefore central to benefit-sharing outcomes.

One of the IDN's current case studies involves the Yirrkala Literature Production Centre in Northeast Arnhem Land (Northern Territory, Australia). Co-designed on Country, the program prioritises governance-visible catalogue records, community-directed access pathways, and secure storage on Yolŋu Country. Authority over access and reuse is embedded directly into descriptions and workflows, enabling cultural continuity, community use, and selective external engagement on Yolŋu terms (Smith & Clarke, 2025). This illustrates how benefit arises when governance is embedded at the point of curation and access, not added downstream.

Q10. Which governance arrangements (for example, standardization, finance, intellectual property and data protection in a broader sense) impact benefits of data and can such arrangements help address equities related to benefit sharing?

Governance arrangements that directly shape benefit-sharing include standardised roles and responsibilities across agents and intermediaries, machine-readable governance rules, contractual arrangements recognising collective rights, and sustained financing of Indigenous-led repositories and infrastructures. Together, these mechanisms shift benefit-sharing from discretionary goodwill to accountable practice, supporting equity across institutional and national settings (Lin et al., 2020; Desai et al., 2016).

Q11. How can investments in digital public goods and infrastructure expand the circle of actors able to derive value from data and participate in data-driven innovation?

Targeted investment in digitally governed public infrastructure, such as interoperable catalogues, reference data resources, trusted or federated environments, and Indigenous-controlled repositories, can expand participation by enabling Indigenous communities and institutions to govern access, derive value, and determine terms of reuse. When designed to embed Indigenous governance rather than extract data, such infrastructures enable participation in data-driven innovation without requiring communities to relinquish authority or control.

Q12. Are there any aspects of this subject that require further studies, standardization, or other empirical work, from a development perspective, that would facilitate our work on the subject?



Approaches to evaluating benefit-sharing should include collective benefit indicators, governance maturity models, comparative evaluations across projects, and reference implementations supported by validation tools. These mechanisms allow benefits to be assessed transparently and over time, rather than assumed, and support learning across jurisdictions and sectors (W3C, 2024; Iannella et al., 2018; Wilkinson et al., 2016).

Q13. Do you believe there are other high-level questions that would add value to Track 3 discussions and are not covered above? If so, please list them.

Key additional considerations include mechanisms for recognising and remedying collective harms, and the requirement for evidence of community-defined benefit in cross-border and international data reuse. Without such evidence, benefit-sharing risks becoming symbolic rather than substantive, particularly in global data ecosystems (EMRIP, 2025).

In international contexts, benefit-sharing should require demonstrable linkage between governance arrangements and community-defined outcomes, rather than assumed global public benefit.

Track 4. Data Flows

Q6. What domestic, regional and international measures currently support cross-border data flows?

Existing measures that support trusted cross-border data flows increasingly rely on governance portability rather than unrestricted transfer. The Australia–Makassar partnership provides a practical example of cross-border governance in operation. In this context, shared repositories, interoperable metadata, and aligned policies enable Indigenous communities of origin and Indigenous knowledge holders to remain referenced wherever materials circulate, with governance conditions and policy links carried alongside records rather than separated from them (Smith & Clarke, 2025).

More broadly, trusted cross-border data flows are supported by combinations of persistent identifiers, interoperable metadata that preserve governance fields, Indigenous intermediaries, and contractual clauses that carry free, prior and informed consent (FPIC), provenance, and cultural protocols across jurisdictions. Technical mechanisms such as ODRL-style rights vocabularies, dashboards, and audit trails further support accountability, while participation in international standards fora and UN-hosted repository governance initiatives provides alignment and oversight at multilateral levels (W3C, 2024; Iannella et al., 2018; Lin et al., 2020).

Q7. What challenges do countries, in particular developing countries, face regarding cross-border data flows?

Countries face a combination of structural and institutional challenges in managing cross-border data flows. These include limited regulatory and technical capacity to govern complex data exchanges;



unequal bargaining power in negotiations with larger states and global technology firms; and insufficient digital infrastructure to support trusted, governed data environments. For Indigenous data, these challenges are compounded by the absence of mechanisms that recognise Indigenous authority once data moves beyond national or institutional boundaries.

Q8. What mechanisms, including data intermediaries, as well as regulations, and international standards exist to promote safe, secure, and trusted data sharing, and how do they interact with developmental benefits?

Mechanisms that promote safe and trusted data sharing increasingly emphasise governance-first approaches. These include the use of Indigenous and public-interest data intermediaries; federated or trusted data environments that bring computation to data rather than exporting sensitive datasets; and standards-based metadata frameworks that encode authority, consent conditions, sensitivity, and permitted or prohibited uses.

When aligned with Indigenous governance, such mechanisms support developmental benefits by enabling data to be used for health, education, environmental management, and cultural preservation without detaching it from community authority. Developmental value is therefore generated through controlled access, accountability, and reciprocity, rather than through unrestricted data extraction or transfer.

Q9. How do these mechanisms ensure safe and trusted data sharing in critical areas like healthcare, emerging technologies, and disaster prevention?

In critical sectors, governance-first data sharing reduces risk while enabling collaboration. In healthcare and population data, trusted environments and governed linkage allow analysis to occur without exposing identifiable or culturally sensitive data to inappropriate reuse. In emerging technologies, including artificial intelligence, machine-readable governance rules and auditable permissions constrain training and reuse to community-approved purposes, mitigating risks of cultural misappropriation and collective harm. In disaster prevention and environmental monitoring, federated data models enable timely data use across borders while preserving Indigenous authority over sensitive ecological and cultural knowledge.

Across these domains, the effectiveness of data sharing depends less on the volume of data transferred and more on the strength of governance embedded within the data flow.

Q10. Are there any other points, ideas, or issues you would like to raise that were not addressed?

Cross-border data flows must not be treated as events that suspend Indigenous governance. When data crosses jurisdictions, Indigenous authority, consent conditions, and accountability obligations should persist. This requires international data governance frameworks to recognise Indigenous



Peoples as continuing rights-holders and governors of their data, and to support mechanisms that make governance portable, visible, and enforceable across borders. Without such measures, cross-border data flows risk reproducing extractive and inequitable practices under the guise of global cooperation.

Preventing collective and cultural harms in cross-border data flows requires governance mechanisms that persist beyond jurisdictional boundaries, rather than relying on post-hoc ethical review or voluntary compliance.



Recommendations to the UN CSTD Working Group on Data Governance

The Indigenous Data Network recommends that the Working Group:

1. **Treat Indigenous Data Governance as a cross-cutting governance layer** in its final framework, rather than as a sectoral, ethical, or population-specific add-on, recognising its relevance across all domains of data governance.
2. **Explicitly recognise Indigenous Peoples as governing authorities** within data systems, with decision-making roles over data relating to them, rather than positioning Indigenous Peoples solely as stakeholders or vulnerable groups.
3. **Recommend that interoperability frameworks require governance-capable metadata**, including the ability to express collective authority, consent conditions, sensitivity, and permitted or prohibited uses, so governance remains visible and auditable as data moves across systems and borders.
4. **Acknowledge the limits of individual consent and privacy-only models**, and incorporate principles and mechanisms capable of addressing collective, intergenerational, and cultural dimensions of data governance.
5. **Identify recognition of Indigenous authority, collective and ongoing FPIC, refusal rights, and auditable accountability as minimum requirements** for rights-based data governance, rather than optional best practice.
6. **Encourage Member States and international organisations to invest in Indigenous-led data infrastructure**, including catalogues, registries, and trusted or federated environments, that enable governance-first data sharing and participation in global data ecosystems without extraction.
7. **Support the integration of Indigenous Data Governance into international standards-setting and data infrastructure initiatives**, so governance principles are reflected in system design, not confined to policy statements.

Without explicit recognition of Indigenous Data Governance, global data governance frameworks risk reproducing extractive and inequitable practices. The measures outlined above provide a practicable pathway for the Working Group to address this gap.



References

- Carroll, S. R., Garba, I., Figueroa-Rodríguez, O. L., Holbrook, J., Lovett, R., Materechera, S., Parsons, M., et al. (2020). The CARE Principles for Indigenous Data Governance. *Data Science Journal*, 19(1), 43. <https://doi.org/10.5334/dsj-2020-043>
- Desai, T., Ritchie, F., & Welpton, R. (2016). Five Safes: Designing data access for research. *Economics Working Paper Series 1601*. University of the West of England, Bristol. <https://www2.uwe.ac.uk/faculties/BBS/Documents/1601.pdf>
- Expert Mechanism on the Rights of Indigenous Peoples (EMRIP). (2025, August 6). *A/HRC/60/66: Right of Indigenous Peoples to data, including with regard to data collection and disaggregation*. Office of the High Commissioner for Human Rights. <https://www.ohchr.org/en/documents/thematic-reports/ahrc6066-right-indigenous-peoples-data-including-regard-data-collection>
- Iannella, R., Steidl, M., Myles, S., & Rodríguez-Doncel, V. (2018, February 15). *ODRL Vocabulary & Expression 2.2* (W3C Recommendation). World Wide Web Consortium. <https://www.w3.org/TR/odrl-vocab/>
- Indigenous Data Network. (2025). IDN Themes vocabulary. The University of Melbourne. <https://data.idnau.org/pid/vocab/idn-th>
- Indigenous Data Network. (2025). IDN Catalogue Profile specification. The University of Melbourne. <https://data.idnau.org/pid/cp/spec>
- Indigenous Data Network. (2024, April 4). *IDN Roundtable: Developing & Adopting Indigenous Data Governance Frameworks [Defining Indigenous Data]*. Melbourne, Victoria, Australia.
- Kukutai, T., & Taylor, J. (Eds.). (2016). *Indigenous data sovereignty: Toward an agenda*. ANU Press. <https://doi.org/10.22459/CAEPR38.11.2016>
- Lin, D., Crabtree, J., Dillo, I., Downs, R. R., Edmunds, R., Giaretta, D., De Giusti, M., et al. (2020). The TRUST Principles for digital repositories. *Scientific Data*, 7*, 144. <https://doi.org/10.1038/s41597-020-0486-7>
- National Indigenous Australians Agency. (2023). *Framework for the governance of Indigenous data*. Australian Government. Canberra, Australia. <https://www.niaa.gov.au/resource-centre/framework-governance-indigenous-data>
- Rose, J., Langton, M., Smith, K., & Clinch, D. (2023). Indigenous Data Governance in Australia: Towards a national framework. *The International Indigenous Policy Journal*, 14(1), 1–30. <https://doi.org/10.18584/iipj.2023.14.1.10987>
- Smith, K., & Clarke, A. (2025, November 27). Current Data Governance Models lack clear recognition of Indigenous jurisdiction over data. *Caring for Indigenous data in museums and research*, Musée du quai Branly – Jacques Chirac, Paris.



United Nations General Assembly. (2007). *United Nations Declaration on the Rights of Indigenous Peoples* (A/RES/61/295). <https://documents.un.org/doc/undoc/gen/n06/512/07/pdf/n0651207.pdf>

Wilkinson, M. D., Dumontier, M., Aalbersberg, I. J., Appleton, G., Axton, M., Baak, A., Blomberg, N., et al. (2016). The FAIR guiding principles for scientific data management and stewardship. *Scientific Data*, 3, 160018. <https://doi.org/10.1038/sdata.2016.18>

World Health Organization. (2021). WHO guideline on data integrity. *Annex 4, WHO Technical Report Series*, No. 1033. <https://www.who.int/publications/m/item/annex-4-trs-1033>

World Wide Web Consortium. (2024, August 22). *Data Catalog Vocabulary (DCAT) — Version 3* (W3C Recommendation). <https://www.w3.org/TR/vocab-dcat-3/>